

Deep Learning

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Multi-layered neural networks that learn complex patterns

Deep Learning uses artificial neural networks with many layers to learn complex patterns directly from raw data, driving major breakthroughs in image recognition, language processing, and more.

● DEFINITION

A subset of machine learning using multi-layered artificial neural networks to model and understand complex patterns in data.

● WHY IT MATTERS

Deep learning has enabled dramatic performance leaps in tasks like image recognition, language translation, and speech processing that were previously very difficult for computers.

● WORKING PRINCIPLE



Deep neural networks process data through many interconnected layers, with each layer learning increasingly abstract features, adjusting millions of internal weights through a training process called backpropagation.

● QUICK FACTS

- ▶ The 'transformer' architecture behind modern language models was introduced in 2017
- ▶ Some deep learning models have hundreds of billions of parameters

● KEY TERMS

- Neural network
- Backpropagation
- Convolutional network
- Transformer

● CORE CONCEPTS

- ▶ Neural network layers
- ▶ Backpropagation
- ▶ Convolutional neural networks
- ▶ Transformer architecture

● APPLICATIONS

- ▶ Image and facial recognition
- ▶ Natural language processing
- ▶ Autonomous vehicle perception
- ▶ Medical image diagnosis

● ADVANTAGES

- ▶ Achieves state-of-the-art performance on complex tasks
- ▶ Automatically learns relevant features from raw data
- ▶ Scales well with more data and computing power

CAREER OPPORTUNITIES

Deep learning engineer, Computer vision engineer, NLP engineer, AI research scientist

RESEARCH AREAS

Transformer architectures, Efficient/low-power neural networks, Multimodal learning

● CURRENT TRENDS

Rapid growth of large language models and multimodal AI systems that combine text, image, and audio understanding.

● REAL-LIFE EXAMPLES

- ▶ Face recognition unlocking smartphones
- ▶ Voice assistants understanding speech
- ▶ Self-driving car object detection

● LIMITATIONS

- ▶ Requires massive datasets and computing resources
- ▶ Difficult to interpret why a model makes specific decisions
- ▶ Training large models consumes significant energy

INDUSTRY APPLICATIONS

Technology, Healthcare imaging, Automotive, Finance

FUTURE SCOPE

Continued growth of large-scale foundation models is expected to further expand deep learning capabilities across new domains.

● SUMMARY

Deep learning uses multi-layered neural networks to learn complex patterns directly from data, driving today's most powerful AI systems.